

Solving Inequalities in Real-World Contexts

Worksheet 7A.01.10

Name _____

Date _____

Period _____

Quinn solved the inequality $2.5 \geq (c - 3)/4$ and made an error. Her work is shown. Use it for questions 1 through 3.

Quinn's work

$$2.5 \geq (c - 3) / 4$$

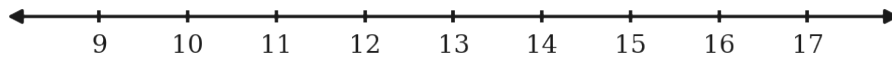
$$2.5 + 3 \geq (c - 3 + 3) / 4$$

$$5.5 \geq c / 4$$

$$22 \geq c$$

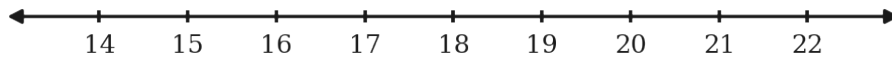
1. What error did Quinn make?
2. Write a brief note to Quinn explaining how to avoid this in the future. Say what she did correctly as well.

3. Solve Quinn's inequality correctly and represent the solution on the number line.

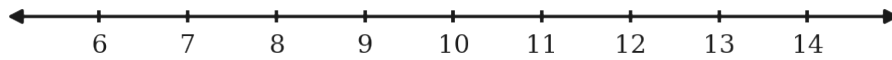


For questions 4 through 7, solve the inequality and represent the solution on the number line.

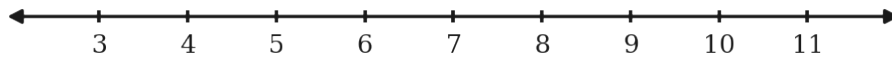
4. $-\frac{b}{4} + 5 > \frac{1}{2}$



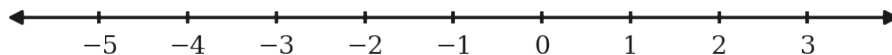
5. $8.8 \geq 0.8(-17.3 + 2.9x)$



6. $1.5m - 10 < -\frac{1}{4}$



7. $-\frac{2}{3}(-3 - 12a) \leq -6$



Mary is buying donuts and juice for her son's third birthday party at school. She has at most \$25 to spend. She bought 10 donuts at \$1.25 each, and each case of juice costs \$3.49. Mary wrote $10(1.25) + 3.49c \leq 25$, where c is the number of cases of juice.

8. Why did Mary use $\leq$ rather than $<$, $>$ or $\geq$?

9. Solve the inequality.

10. Will Mary be able to purchase 4 cases of juice? Justify your answer mathematically.